

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.

- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network Next Hop

192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.

- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

Question 1

Given Configuration

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

Step 1: Identify the Subnet

A /26 subnet has a block size of **64**.

Subnets are:

- 192.168.10.0 – 192.168.10.63
- **192.168.10.64 – 192.168.10.127**
- 192.168.10.128 – 192.168.10.191
- 192.168.10.192 – 192.168.10.255

Since **192.168.10.65** lies between **64–127**, it belongs to:

Network Address: 192.168.10.64/26

Step 2: Network Information

Item	Value
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Valid Host Range	192.168.10.65 – 192.168.10.126

Step 3: Verify IP and Gateway

- Workstation IP **192.168.10.65** → ✓ Valid
- Gateway **192.168.10.1** → ✗ Invalid (belongs to another subnet)

Step 4: Configuration Error

The default gateway is in **192.168.10.0/26**, while the workstation is in **192.168.10.64/26**.

Step 5: Optimized Configuration

Parameter	Correct Value
IP Address	192.168.10.66 (or any 65–126)
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.65 (router interface)

Step 6: Usable Hosts

Host bits = 6

Total addresses = $2^6 = 64$

Usable hosts = **64 – 2 = 62**

Final Answer

- Network = **192.168.10.64/26**
- Broadcast = **192.168.10.127**
- Host Range = **192.168.10.65–192.168.10.126**
- Error = Incorrect gateway
- Correct Gateway = **192.168.10.65**
- Usable Hosts = **62**

Question 2

Given IPv6 Addresses

Device A

2001:db8::ff00:42:8329/64

Device B

2001:db8:0:0:1::1/64

Step 1: Expanded Addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network Prefix

With /64, the first four blocks form the network.

Device A

2001:0db8:0000:0000::/64

Device B

2001:0db8:0000:0000::/64

Step 3: Same Subnet?

✓ Yes.

Both belong to

2001:db8:0:0::/64

Step 4: Possible Issue

IPv6 addresses are correct.

Communication failure may be due to:

- IPv6 disabled
- Wrong default gateway
- Router Advertisement unavailable
- Firewall blocking ICMPv6

Step 5: Optimized Addresses

Device	Address
--------	---------

Device A	2001:db8:0:0::10/64
----------	---------------------

Device B	2001:db8:0:0::20/64
----------	---------------------

Step 6: Hosts

Host bits = 64

Available addresses

$2^{64} = 18,446,744,073,709,551,616$

Question 3

Given Network

192.168.1.0/24

Subnets

- /26
- /27
- /28

Current Allocation

Subnet	Network	Host Range	Broadcast	Usable Hosts
/26	192.168.1.0	1–62	63	62
/27	192.168.1.64	65–94	95	30
/28	192.168.1.96	97–110	111	14

Analysis

Subnet	Usable Hosts	Observation
/26	62	Large department
/27	30	Medium department
/28	14	Small department

Some departments have unused IP addresses, while others may require additional hosts.

Debugging

Problem:

Fixed subnet allocation causes inefficient utilization.

Optimized Solution (VLSM)

Assign larger subnets only where needed.

Example:

Department Required Hosts Recommended Subnet

Dept A	50	/26
Dept B	25	/27
Dept C	10	/28

VLSM reduces IP address wastage while meeting host requirements.

Question 4

Given IPv6 Addresses

Device A

2001:db8::ff00:42:8329/64

Device B

2001:db8:0:0:1::1/64

Expanded Form

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix

Both:

2001:0db8:0000:0000::/64

Verification

✓ Same subnet

Optimization

Assign

- Device A → **2001:db8::10/64**
- Device B → **2001:db8::20/64**

Ensure:

- IPv6 routing enabled
- Same gateway
- Router Advertisement configured

Total Hosts

$$2^{64} = 18,446,744,073,709,551,616$$

Question 5

Routing Table

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP

192.168.2.100

Step 1: Longest Prefix Match

Destination belongs to

192.168.2.0/24

Step 2: Matching Entry

Destination	Next Hop
192.168.2.0/24	10.0.0.3

Step 3: Next-Hop Router

Packet forwarded to

10.0.0.3

Step 4: Verify Destination

Destination

192.168.2.100

Host range of subnet:

192.168.2.1 – 192.168.2.254

✓ Valid

Step 5: Possible Errors

If packets fail:

- Incorrect next-hop configuration
- Interface shutdown
- Missing route
- Wrong subnet mask
- ACL blocking traffic

Step 6: Optimization

- Verify next-hop **10.0.0.3** is reachable.
- Configure route summarization where appropriate.
- Remove unnecessary routing entries.
- Verify interfaces are up (`no shutdown`).

Step 7: Default Route

If destination does not match any routing entry,

Use:

Default Route → 10.0.0.1